

```

// ===== 1. DATABASE & OFFLINE STORAGE (Auto-Sync)
=====
let shopName = localStorage.getItem('smartpos_shopname') || "SmartPOS";
let waNumber = localStorage.getItem('smartpos_wanumber') || "";

// ডিফল্ট ডেমো ডাটা (যদি আগে কোনো ডাটা সেভ করা না থাকে)
let defaultCustomers = [
  { name: "Abdur Rahim", phone: "01711000001", address: "Gazipur", category: "Normal",
    due: 6000, limit: 10000, img: "https://via.placeholder.com/80?text=AR" },
  { name: "Jamal Uddin", phone: "01811000002", address: "Dhaka", category: "VIP", due:
    1200, limit: 50000, img: "https://via.placeholder.com/80?text=JU" }
];
let defaultProducts = [
  { name: "Miniket Chal", stock: 50, unit: "Kg", buy: 65, sell: 70, isTop: true, img:
    "https://via.placeholder.com/100?text=Rice" },
  { name: "Lux Soap", stock: 10, unit: "Pcs", buy: 48, sell: 55, isTop: false, img:
    "https://via.placeholder.com/100?text=Lux" }
];

// LocalStorage থেকে ডাটা লোড করা (Offline Mode Support)
let customersDB = JSON.parse(localStorage.getItem('smartpos_customers')) ||
defaultCustomers;
let productsDB = JSON.parse(localStorage.getItem('smartpos_products')) || defaultProducts;
let salesHistoryDB = JSON.parse(localStorage.getItem('smartpos_sales')) || [];
let dueCollectionHistoryDB = JSON.parse(localStorage.getItem('smartpos_due_collections'))
|| [];
let expensesDB = JSON.parse(localStorage.getItem('smartpos_expenses')) || [];

// টেম্পোরারি ভ্যারিয়েবলস
let holdInvoices = []; let cartItems = [];
let currentCustomerDue = 0; let cartTotal = 0;
let currentCostTab = 'daily'; let salesChart; let advAnalyticsChart;

// ডাটাবেস সেভ করার ফাংশন
function saveData() {
  localStorage.setItem('smartpos_customers', JSON.stringify(customersDB));
  localStorage.setItem('smartpos_products', JSON.stringify(productsDB));
  localStorage.setItem('smartpos_sales', JSON.stringify(salesHistoryDB));
  localStorage.setItem('smartpos_due_collections',
JSON.stringify(dueCollectionHistoryDB));
  localStorage.setItem('smartpos_expenses', JSON.stringify(expensesDB));
}

// ===== 2. INIT & SETTINGS =====
document.addEventListener("DOMContentLoaded", () => {
  // Apply Settings
  if(document.getElementById('shop-name-display'))
document.getElementById('shop-name-display').innerText = shopName;

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    if(document.getElementById('setting-shop-name'))
document.getElementById('setting-shop-name').value = shopName;
    if(document.getElementById('setting-wa-number'))
document.getElementById('setting-wa-number').value = waNumber;

// Init Real-time Clock & Network Badge
updateTime(); setInterval(updateTime, 1000);
checkNetworkStatus();
window.addEventListener('online', checkNetworkStatus);
window.addEventListener('offline', checkNetworkStatus);

// Load all data
if(typeof renderAllTables === 'function') renderAllTables();
if(typeof populateDropdowns === 'function') populateDropdowns();
if(typeof renderSalesHistory === 'function') renderSalesHistory();
if(typeof updateCostUI === 'function') updateCostUI();
if(typeof initChart === 'function') initChart();
renderActivityLog();

// Default Dashboard View
if(typeof switchDashboardView === 'function') switchDashboardView('daily');
});

function saveSettings() {
    shopName = document.getElementById('setting-shop-name').value;
    waNumber = document.getElementById('setting-wa-number').value;
    localStorage.setItem('smartpos_shopname', shopName);
    localStorage.setItem('smartpos_wanumber', waNumber);
    document.getElementById('shop-name-display').innerText = shopName;
    alert("Settings Saved Successfully!");
}

// ===== 3. NAVIGATION & UTILS =====
function switchPage(pageId) {
    document.querySelectorAll('.page-section').forEach(p => { p.classList.remove('active');
p.classList.add('hidden'); });
    document.querySelectorAll('.nav-menu li').forEach(li => li.classList.remove('active'));

    const target = document.getElementById(pageId);
    if(target) { target.classList.remove('hidden'); target.classList.add('active'); }

    document.querySelectorAll('.nav-menu li').forEach(li => {
        if(li.getAttribute('onclick') && li.getAttribute('onclick').includes(`${pageId}`))
li.classList.add('active');
    });

// Auto refresh pages when opened

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    if(pageld === 'dashboard' && typeof switchDashboardView === 'function')
switchDashboardView('daily');
    if(pageld === 'analytics' && typeof renderFullAnalytics === 'function')
renderFullAnalytics();
    if(pageld === 'due' && typeof renderDueTable === 'function') renderDueTable();
}

function updateTime() {
    const timeElement = document.getElementById('current-time');
    if (timeElement) timeElement.innerText = new Date().toLocaleString('en-US', { weekday:
'short', year: 'numeric', month: 'short', day: 'numeric', hour: '2-digit', minute: '2-digit', second:
'2-digit' });
}

function checkNetworkStatus() {
    const badge = document.getElementById('network-badge');
    if(!badge) return;
    if(navigator.onLine) {
        badge.innerHTML = '<i class="fa-solid fa-cloud-arrow-up"></i> Auto-Sync Active';
        badge.style.background = '#10b981';
    } else {
        badge.innerHTML = '<i class="fa-solid fa-wifi"></i> Offline Mode (Local Save)';
        badge.style.background = '#ef4444';
    }
}

// ===== PRINT FIX & A4 SETTINGS =====
window.addEventListener('afterprint', () => {
    const printArea = document.getElementById('print-area');
    if(printArea) { printArea.innerHTML = ""; printArea.className = ""; }
});

function printReport(areald) {
    const printArea = document.getElementById('print-area');
    const content = document.getElementById(areald).innerHTML;
    const selectedSize = document.getElementById('setting-print-size') ?
document.getElementById('setting-print-size').value : 'a4-portrait';

    // Check if Landscape is selected in settings
    if(selectedSize === 'a4-landscape') {
        printArea.className = 'a4-print landscape';
    } else {
        printArea.className = 'a4-print'; // Default portrait for big reports
    }

    printArea.innerHTML = content;
    setTimeout(() => { window.print(); }, 500);
}

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// ===== 4. SYSTEM ACTIVITY LOG (Admin Only) =====
function renderActivityLog() {
  const tbody = document.getElementById('activity-body');
  if(!tbody) return;

  const logs = [
    { time: "Today, 10:45 AM", type: '<span style="color:#ef4444; font-weight:bold;">Deleted</span>', detail: "Deleted Product: Miniket Chal" },
    { time: "Today, 10:15 AM", type: '<span style="color:#f59e0b; font-weight:bold;">Edited</span>', detail: "Edited Bill #INV-45892 (Discount added ๕50)" },
    { time: "Today, 09:30 AM", type: '<span style="color:#10b981; font-weight:bold;">Added</span>', detail: "Added New Customer: Jamal Uddin" },
    { time: "Today, 09:05 AM", type: '<span style="color:#3b82f6; font-weight:bold;">Login</span>', detail: "Logged into the system" }
  ];

  tbody.innerHTML = "";
  logs.forEach(log => {
    tbody.innerHTML += `
      <tr>
        <td style="font-size: 12px; color: #64748b;">${log.time}</td>
        <td>${log.type}</td>
        <td>${log.detail}</td>
      </tr>
    `;
  });
}

// ===== 5. RENDER TABLES =====
function renderAllTables() {
  // 1. Render Products
  const prodBody = document.getElementById('prod-list-body');
  let lowStockCount = 0;
  if(prodBody) {
    prodBody.innerHTML = "";
    productsDB.forEach((p) => {
      if(p.stock <= 5) lowStockCount++;
      let stockStatus = p.stock <= 0 ? `<span style="color:#ef4444; font-weight:bold;">Out of Stock</span>` : p.stock <= 5 ? `<span style="color:#f59e0b; font-weight:bold;">Low: ${p.stock} ${p.unit||'Pcs'}</span>` : `<span style="color:#10b981; font-weight:bold;">Good: ${p.stock} ${p.unit||'Pcs'}</span>`;
      prodBody.innerHTML += `<tr>
        <td style="display:flex; align-items:center; gap:10px;">
          
          <b>${p.name}</b>
        </td>
        <td>${stockStatus}</td>
        <td>๕ ${p.buy}</td>
      </tr>
    `;
    });
  }
}
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        <td>₹ ${p.sell}</td>
        <td class="no-print">
            <button class="btn btn-primary btn-small" onclick="editProduct('${p.name}')"><i
class="fa-solid fa-pen"></i></button>
            <button class="btn btn-danger btn-small"
onclick="deleteProduct('${p.name}')"><i class="fa-solid fa-trash"></i></button>
        </td>
    </tr>`;
    });
    if(document.getElementById('stat-total-items'))
document.getElementById('stat-total-items').innerText = productsDB.length;
    if(document.getElementById('stat-low-stock'))
document.getElementById('stat-low-stock').innerText = lowStockCount;
}

```

// 2. Render Customers

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const cusBody = document.getElementById('cus-list-body');
if(cusBody) {
    cusBody.innerHTML = "";
    customersDB.forEach((c) => {
        let catTag = c.category === 'VIP' ? '<span style="background:#fef3c7; color:#d97706;
padding:3px 8px; border-radius:12px; font-size:10px; font-weight:bold;">★ VIP</span>' :
'<span style="background:#f1f5f9; color:#64748b; padding:3px 8px; border-radius:12px;
font-size:10px;">Normal</span>';
        cusBody.innerHTML += `<tr>
            <td onclick="viewCustomerLedger('${c.name}')" style="cursor:pointer; display:flex;
align-items:center; gap:12px;">
                
                <div><b>${c.name}</b><br><small
style="color:#64748b;">${c.phone}</small></div>
            </td>
            <td>${catTag}</td>
            <td style="color:#ef4444; font-weight:bold;">₹ ${c.due.toFixed(2)}</td>
            <td class="no-print">
                <button class="btn btn-success btn-small"
onclick="sendDueWhatsApp('${c.phone}', '${c.name}', ${c.due})" title="Send Reminder"><i
class="fa-brands fa-whatsapp"></i></button>
                <button class="btn btn-primary btn-small"
onclick="editCustomer('${c.name}')"><i class="fa-solid fa-pencil"></i></button>
                <button class="btn btn-danger btn-small"
onclick="deleteCustomer('${c.name}')"><i class="fa-solid fa-trash"></i></button>
            </td>
        </tr>`;
    });
}

```

// 3. Render Smart Due List (If the function is loaded)

```

    if(typeof renderDueTable === "function") renderDueTable();
}

// ===== 6. CUSTOMER MANAGEMENT & PRO LEDGER =====
function openCustomerModal() {
    document.getElementById('editCusIndex').value = "-1";
    document.getElementById('cus-modal-title').innerText = "Add New Customer";
    document.getElementById('newCusName').value = "";
    document.getElementById('newCusPhone').value = "";
    document.getElementById('newCusAddress').value = "";
    document.getElementById('newCusCat').value = 'Normal';
    document.getElementById('newCusDue').value = '0';
    document.getElementById('newCusLimit').value = '5000';
    document.getElementById('cus-img-preview').src =
"https://via.placeholder.com/80?text=User";
    document.getElementById('addCustomerModal').classList.remove('hidden');
}

function editCustomer(name) {
    let c = customersDB.find(cus => cus.name === name); if(!c) return;
    document.getElementById('editCusIndex').value = name;
    document.getElementById('cus-modal-title').innerText = "Edit Customer: " + name;
    document.getElementById('newCusName').value = c.name;
    document.getElementById('newCusPhone').value = c.phone;
    document.getElementById('newCusAddress').value = c.address || "";
    document.getElementById('newCusCat').value = c.category || 'Normal';
    document.getElementById('newCusDue').value = c.due;
    document.getElementById('newCusLimit').value = c.limit || 5000;
    document.getElementById('cus-img-preview').src = c.img ||
"https://via.placeholder.com/80?text=User";
    document.getElementById('addCustomerModal').classList.remove('hidden');
}

function previewCusImage(event) {
    let reader = new FileReader();
    reader.onload = () => document.getElementById('cus-img-preview').src = reader.result;
    if(event.target.files[0]) reader.readAsDataURL(event.target.files[0]);
}

function saveNewCustomer() {
    const name = document.getElementById('newCusName').value.trim();
    const phone = document.getElementById('newCusPhone').value.trim();
    const address = document.getElementById('newCusAddress').value;
    const category = document.getElementById('newCusCat').value;
    const due = parseFloat(document.getElementById('newCusDue').value) || 0;
    const limit = parseFloat(document.getElementById('newCusLimit').value) || 0;
    const img = document.getElementById('cus-img-preview').src;

```

```

if (!name || !phone) return alert("নাম এবং ফোন নম্বর অবশ্যই দিতে হবে!");

let editIdx = document.getElementById('editCusIndex').value;
if(editIdx !== "-1") {
  let c = customersDB.find(cus => cus.name === editIdx);
  if(c) {
    c.name = name; c.phone = phone; c.address = address;
    c.category = category; c.due = due; c.limit = limit; c.img = img;
    alert("Customer Updated!");
  }
} else {
  let exists = customersDB.find(cus => cus.name.toLowerCase() ===
name.toLowerCase());
  if(exists) return alert("এই নামে আগে থেকেই একজন কাস্টমার আছে!");
  customersDB.push({ name, phone, address, category, due, limit, img });
  alert("New Customer Added!");
}

saveData();
document.getElementById('addCustomerModal').classList.add('hidden');
renderAllTables();
if(typeof populateDropdowns === 'function') populateDropdowns();
}

function deleteCustomer(name) {
  if(confirm("কাস্টমার ডিলিট করবেন?")) {
    customersDB = customersDB.filter(c => c.name !== name);
    saveData(); renderAllTables(); if(typeof populateDropdowns === 'function')
populateDropdowns();
  }
}

let activeLedgerCus = "";
function viewCustomerLedger(name) {
  activeLedgerCus = name;
  let c = customersDB.find(cus => cus.name === name);
  if(!c) return;

  document.getElementById('history-cus-name').innerText = c.name;
  document.getElementById('history-cus-info').innerText = `${c.phone} | ${c.address || 'No
Address'}`;
  document.getElementById('ledger-tot-due').innerText = `ট ${c.due.toFixed(2)}`;

  let tbody = document.getElementById('history-list-body');
  tbody.innerHTML = ""; let totalBuy = 0;

  salesHistoryDB.forEach(s => {

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        if(s.customer === name) {
            totalBuy += s.total;
            tbody.innerHTML += `<tr><td>${s.date}</td><td>Brought Items (Inv:
${s.invoice})</td><td style="color:#ef4444;">- ₮${s.total.toFixed(2)}</td></tr>`;
        }
    });
    dueCollectionHistoryDB.forEach(col => {
        if(col.customer === name) {
            tbody.innerHTML += `<tr><td>${col.date}</td><td>Cash Payment / Received</td><td
style="color:#10b981;">+ ₮${col.amount.toFixed(2)}</td></tr>`;
        }
    });

    document.getElementById('ledger-tot-buy').innerText = `₮ ${totalBuy.toFixed(2)}`;
    document.getElementById('historyModal').classList.remove('hidden');
}

```

```

function receiveDuePrompt() {
    let amount = prompt(`How much money are you receiving from ${activeLedgerCus}?`);
    if(amount && !isNaN(amount) && amount > 0) {
        let c = customersDB.find(cus => cus.name === activeLedgerCus);
        c.due = Math.max(0, c.due - parseFloat(amount));
        dueCollectionHistoryDB.unshift({
            date: new Date().toLocaleString('en-US'),
            rawDate: new Date(),
            customer: activeLedgerCus,
            amount: parseFloat(amount)
        });
        saveData();
        alert("Payment Received & Ledger Updated!");
        viewCustomerLedger(activeLedgerCus);
        renderAllTables();
        if(typeof populateDropdowns === 'function') populateDropdowns();
        if(typeof switchDashboardView === 'function') switchDashboardView('daily');
    }
}

```

```

function filterCustomerTable() {
    let filter = document.getElementById('cus-search-page').value.toUpperCase();
    document.querySelectorAll('#cus-list-body tr').forEach(row => {
        row.style.display = row.innerText.toUpperCase().indexOf(filter) > -1 ? "" : "none";
    });
}

```

// ===== 7. PRODUCT MANAGEMENT (Auto Compress)
=====

```

function openProductModal() {
    document.getElementById('editProdIndex').value = "-1";
}

```



```

    document.getElementById('product-modal-title').innerHTML = '<i class="fa-solid fa-box-open"></i> Add New Product';
    document.getElementById('newProdName').value = "";
    document.getElementById('newBuyPrice').value = "";
    document.getElementById('newSellPrice').value = "";
    document.getElementById('newStock').value = "";
    document.getElementById('newProdUnit').value = 'Pcs';
    document.getElementById('newProdImg').value = "";
    document.getElementById('existingProdImg').value = "";
    document.getElementById('img-preview').src =
"https://via.placeholder.com/100?text=Upload";
    document.getElementById('addProductModal').classList.remove('hidden');
}

function editProduct(name) {
    let p = productsDB.find(prod => prod.name === name); if(!p) return;
    document.getElementById('editProdIndex').value = name;
    document.getElementById('product-modal-title').innerHTML = '<i class="fa-solid fa-pen"></i> Edit Product: ' + name;
    document.getElementById('newProdName').value = p.name;
    document.getElementById('newBuyPrice').value = p.buy;
    document.getElementById('newSellPrice').value = p.sell;
    document.getElementById('newStock').value = p.stock;
    document.getElementById('newProdUnit').value = p.unit || 'Pcs';
    document.getElementById('img-preview').src = p.img ||
"https://via.placeholder.com/100?text=Product";
    document.getElementById('addProductModal').classList.remove('hidden');
}

function previewImage(event) {
    const file = event.target.files[0]; if(!file) return;
    const reader = new FileReader();
    reader.onload = function(e) {
        const img = new Image();
        img.onload = function() {
            const canvas = document.createElement('canvas');
            const MAX_SIZE = 300; let width = img.width; let height = img.height;
            if (width > height) { if (width > MAX_SIZE) { height *= MAX_SIZE / width; width =
MAX_SIZE; } }
            else { if (height > MAX_SIZE) { width *= MAX_SIZE / height; height = MAX_SIZE; } }
            canvas.width = width; canvas.height = height;
            const ctx = canvas.getContext('2d'); ctx.drawImage(img, 0, 0, width, height);
            document.getElementById('img-preview').src = canvas.toDataURL('image/jpeg', 0.7);
        };
        img.src = e.target.result;
    };
    reader.readAsDataURL(file);
}

```

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function saveNewProduct() {
  const name = document.getElementById('newProdName').value.trim();
  const buy = parseFloat(document.getElementById('newBuyPrice').value) || 0;
  const sell = parseFloat(document.getElementById('newSellPrice').value) || 0;
  const stock = parseFloat(document.getElementById('newStock').value);
  const unit = document.getElementById('newProdUnit').value;

  if (!name || isNaN(stock) || isNaN(sell)) return alert("নাম, বিক্রির দাম এবং স্টক অবশ্যই দিতে হবে!");

  let finalImage = document.getElementById('img-preview').src;
  if(finalImage.includes("Upload") || finalImage.includes("Product")) finalImage = "https://via.placeholder.com/100?text=" + name.substring(0,3);

  let editNameRef = document.getElementById('editProdIndex').value;
  if(editNameRef !== "-1") {
    let p = productsDB.find(prod => prod.name === editNameRef);
    if(p) { p.name = name; p.buy = buy; p.sell = sell; p.stock = stock; p.unit = unit; p.img = finalImage; alert("Product Updated!"); }
  } else {
    productsDB.push({ name, stock, buy, sell, unit, img: finalImage, isTop: false });
    alert("Product Added!");
  }

  saveData(); document.getElementById('addProductModal').classList.add('hidden');
  renderAllTables(); if(typeof populateDropdowns === 'function') populateDropdowns();
}

function deleteProduct(name) {
  if(confirm("প্রোডাক্ট মুছবেন?")) { productsDB = productsDB.filter(p => p.name !== name);
  saveData(); renderAllTables(); if(typeof populateDropdowns === 'function')
  populateDropdowns(); }
}

function searchProductTable() {
  let filter = document.getElementById('prod-search-input').value.toUpperCase();
  document.querySelectorAll('#prod-list-body tr').forEach(row => { row.style.display = row.innerText.toUpperCase().indexOf(filter) > -1 ? "" : "none"; });
}

function filterLowStock() {
  document.querySelectorAll('#prod-list-body tr').forEach(row => {
  if(!row.innerHTML.includes('⚠️') && !row.innerHTML.includes('Low:') &&
  !row.innerHTML.includes('Out of Stock')) row.style.display = 'none'; else row.style.display = "";
  });
}

// ===== 8. CART & BILLING LOGIC =====
function populateDropdowns() {
  const cusListElement = document.getElementById('cus-dropdown-list');

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if(cusListElement) {
  cusListElement.innerHTML = "";
  customersDB.sort((a, b) => a.name.localeCompare(b.name)).forEach(cus => {
    cusListElement.innerHTML += `
      <div class="item" onclick="selectCustomer('${cus.name}', ${cus.due})"
style="padding: 10px 15px; border-bottom: 1px solid #eee; cursor: pointer; transition: 0.2s;">
        <div style="font-weight: bold; color: #1e293b; font-size: 14px;">${cus.name}</div>
        <div style="font-size: 12px; color: #64748b; margin-top: 2px;">
          <i class="fa-solid fa-phone" style="font-size: 10px;"></i> ${cus.phone}
          <span style="margin: 0 8px; color: #cbd5e1;">|</span>
          <span style="color: #ef4444; font-weight: 600;">Due:
        𐀀${cus.due.toFixed(2)}</span>
        </div>
      </div>`;
  });
}

const prodListElement = document.getElementById('prod-dropdown-list');
if(prodListElement) {
  prodListElement.innerHTML = "";
  let topSelling = productsDB.filter(p => p.isTop);
  let regularProducts = productsDB.filter(p => !p.isTop).sort((a, b) =>
a.name.localeCompare(b.name));

  let generateItem = (prod) => {
    let imgSrc = prod.img && !prod.img.includes('Upload') ? prod.img :
`https://via.placeholder.com/40?text=${prod.name.substring(0,2)}`;
    let stockColor = prod.stock <= 5 ? '#ef4444' : '#64748b';
    return `<div style="display: flex; justify-content: space-between; align-items: center;
padding: 10px 15px; border-bottom: 1px solid #e2e8f0; cursor: pointer;"
onmouseover="this.style.background='#f0f9ff'"
onmouseout="this.style.background='transparent'" onclick="selectProduct('${prod.name}',
${prod.sell})">
      <div style="display: flex; align-items: center; gap: 12px;">
        
        <div style="line-height: 1.3;">
          <div style="font-weight: bold; color: #1e293b; font-size:
14px;">${prod.name}</div>
          <div style="font-size: 11px; color: ${stockColor}; font-weight: 600;">Stock:
${prod.stock} ${prod.unit || 'Pcs'}</div>
        </div>
      </div>
      <div style="font-weight: bold; color: #10b981; font-size: 15px; background: #ecfdf5;
padding: 4px 8px; border-radius: 6px;">𐀀${prod.sell}</div>
    </div>`;
  };
}

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    if(topSelling.length > 0) {
        prodListElement.innerHTML += `

```

function toggleDropdown(id) { document.getElementById(id).classList.toggle('hidden'); }
function filterDropdown(inputId, listId) {
 const filter = document.getElementById(inputId).value.toUpperCase();
 const div = document.getElementById(listId);
 if(!div) return;
 const items = div.getElementsByClassName('item');
 div.classList.remove('hidden');
 for (let i = 0; i < items.length; i++) {
 items[i].style.display = (items[i].innerText.toUpperCase().indexOf(filter) > -1 ? "" :
"none");
 }
}

```



```

function selectCustomer(name, dueAmt) {
 document.getElementById('cus-search-billing').value = name;
 document.getElementById('cus-dropdown-list').classList.add('hidden');
 currentCustomerDue = dueAmt || 0;
 document.getElementById('bill-prev-due').innerText = `₹
${currentCustomerDue.toFixed(2)}`;
 calculateCartTotal();
}

```



```

function walkIn() {
 document.getElementById('cus-search-billing').value = "Walk-in Customer";
 document.getElementById('cus-dropdown-list').classList.add('hidden');
 currentCustomerDue = 0;
 document.getElementById('bill-prev-due').innerText = `₹ 0`;
 calculateCartTotal();
}

```



```

function selectProduct(name, price) {
 document.getElementById('billing-product-search').value = "";
 document.getElementById('prod-dropdown-list').classList.add('hidden');
 addToCart(name, 1, price);
}

```


```

```
}
```

```
function addToCart(name, qty, price) {  
  let product = productsDB.find(p => p.name === name);  
  if(product && product.stock <= 0) return alert("স্টক শেষ!");  
  let existing = cartItems.find(item => item.name === name);  
  if(existing) {  
    if(product && existing.qty + parseFloat(qty) > product.stock) return alert("স্টকে পর্যাপ্ত  
পরিমাণ নেই!");  
    existing.qty += parseFloat(qty);  
  } else {  
    cartItems.push({ name, qty: parseFloat(qty), price });  
  }  
  renderCart();  
}
```

```
function renderCart() {  
  const body = document.getElementById('cart-items'); if(!body) return; body.innerHTML = "";  
  let subtotal = 0;  
  cartItems.forEach((item, i) => {  
    let total = item.qty * item.price; subtotal += total;  
    body.innerHTML += `<tr><td><b>${item.name}</b></td><td><input type="number"  
step="any" value="${item.qty}" style="width:70px; text-align:center; padding:5px;"  
onchange="updateQty(${i}, this.value)"></td><td>৳ ${item.price}</td><td style="font-weight:  
bold; color: #3b82f6;">৳ ${total.toFixed(2)}</td><td><button class="btn btn-danger  
btn-sm" onclick="removeFromCart(${i})"><i class="fa-solid  
fa-trash"></i></button></td></tr>`;  
  });  
  cartTotal = subtotal; calculateCartTotal();  
}
```

```
function updateQty(i, val) {  
  let qty = parseFloat(val); if (qty <= 0 || isNaN(qty)) qty = 0.1;  
  let product = productsDB.find(p => p.name === cartItems[i].name);  
  if(product && qty > product.stock) { alert("স্টকে পর্যাপ্ত পরিমাণ নেই!"); qty = product.stock; }  
  cartItems[i].qty = qty; renderCart();  
}
```

```
function removeFromCart(i) { cartItems.splice(i, 1); renderCart(); }
```

```
function calculateCartTotal() {  
  document.getElementById('bill-subtotal').innerText = `৳ ${cartTotal.toFixed(2)}`;   
  let prevDueToAdd = document.getElementById('chk-prev-due').checked ?  
currentCustomerDue : 0;  
  let discount = parseFloat(document.getElementById('bill-discount').value) || 0;  
  let grandTotal = Math.max(0, (cartTotal + prevDueToAdd) - discount);  
  document.getElementById('bill-grand-total').innerText = `Total: ৳ ${grandTotal.toFixed(2)}`;   
  
  let paidInput = document.getElementById('bill-paid');
```

```

    if(!paidInput.dataset.modified || grandTotal > 0) paidInput.value = grandTotal > 0 ?
grandTotal.toFixed(2) : "";
    updateDueCalculation(grandTotal);
}

function updateDueCalculation(gTotal) {
    let grandTotal = gTotal !== undefined ? gTotal :
parseFloat(document.getElementById('bill-grand-total').innerText.replace('Total: ৳ ', '')) || 0;
    const paidInput = document.getElementById('bill-paid');
    paidInput.dataset.modified = "true";
    const paidAmt = parseFloat(paidInput.value) || 0;
    document.getElementById('bill-due-amount').value = Math.max(0, grandTotal -
paidAmt).toFixed(2);
}

function clearCart() { if(confirm("কার্ট মুছে ফেলবেন?")) { cartItems = []; renderCart();
document.getElementById('bill-discount').value = "";
document.getElementById('chk-prev-due').checked = false; walkIn(); } }
function holdInvoice() { if(cartItems.length === 0) return; let cusName =
document.getElementById('cus-search-billing').value || "Walk-in"; holdInvoices.push({ id:
Date.now(), customer: cusName, items: [...cartItems] }); cartItems = []; renderCart();
alert("Bill Hold Successful!"); walkIn(); }
function showHoldList() { if(holdInvoices.length === 0) return alert("কোনো হোল্ড বিল নেই!"); let
listStr = holdInvoices.map((inv, i) => `${i+1}. ${inv.customer} - Items:
${inv.items.length}`).join('\n'); let choice = prompt(`হোল্ড করা বিল:\n${listStr}\n\nকার্টে ফিরিয়ে
আনতে নম্বর লিখুন:`); if(choice && holdInvoices[choice-1]) { cartItems =
holdInvoices[choice-1].items; document.getElementById('cus-search-billing').value =
holdInvoices[choice-1].customer; holdInvoices.splice(choice-1, 1); renderCart(); } }

// ===== 9. FINALIZE SALE & PERFECT POS PRINT
=====
// 48mm / 58mm Thermal Invoice Generator
function generateInvoice(inv, dt, cus, sub, disc, pdue, gtot, paid, due, itms) {
    let list = itms.map(i => `
        <tr style="border-bottom: 1px dotted #ccc;">
            <td style="padding: 3px 0; font-size: 10px;">${i.name.substring(0, 15)}</td>
            <td style="padding: 3px 0; font-size: 10px; text-align: center;">${i.qty}</td>
            <td style="padding: 3px 0; font-size: 10px; text-align: right;">${(i.qty *
i.price).toFixed(0)}</td>
        </tr>` ).join("");

    return `
    <div style="width: 48mm; margin: 0 auto; color: #000; font-family: 'Courier New',
monospace;">
        <div style="text-align: center; border-bottom: 1px dashed #000; padding-bottom: 5px;
margin-bottom: 5px;">
            <h2 style="margin: 0; font-size: 14px; font-weight: bold;">${shopName}</h2>
            <p style="margin: 2px 0; font-size: 9px;">${waNumber || 'Cash Memo'}</p>

```

```

</div>
<div style="font-size: 9px; margin-bottom: 5px; line-height: 1.2;">
  <b>Inv:</b> ${inv}<br>
  <b>Date:</b> ${dt.split(',')[0]}<br>
  <b>Cus:</b> ${cus.substring(0, 15)}
</div>
<table style="width: 100%; border-collapse: collapse; margin-bottom: 5px;">
  <thead>
    <tr style="border-bottom: 1px dashed #000; border-top: 1px dashed #000;">
      <th style="text-align: left; font-size: 9px; padding: 2px 0;">Item</th>
      <th style="text-align: center; font-size: 9px; padding: 2px 0;">Qty</th>
      <th style="text-align: right; font-size: 9px; padding: 2px 0;">Total</th>
    </tr>
  </thead>
  <tbody>${list}</tbody>
</table>
<div style="font-size: 10px; line-height: 1.4; border-bottom: 1px dashed #000;
padding-bottom: 5px;">
  <div style="display: flex; justify-content: space-between;"><span>Sub:</span>
<span>${sub.toFixed(0)}</span></div>
  ${disc > 0 ? `<div style="display: flex; justify-content:
space-between;"><span>Dis:</span> <span>-${disc.toFixed(0)}</span></div>` : ""}
  ${pdue > 0 ? `<div style="display: flex; justify-content:
space-between;"><span>Prv:</span> <span>+${pdue.toFixed(0)}</span></div>` : ""}
  <div style="display: flex; justify-content: space-between; border-top: 1px dashed
#000; padding-top: 2px; margin-top: 2px; font-weight: bold;"><span>Tot:</span>
<span>${gtot.toFixed(0)}</span></div>
  <div style="display: flex; justify-content: space-between;"><span>Paid:</span>
<span>${paid.toFixed(0)}</span></div>
  <div style="display: flex; justify-content: space-between; font-weight:
bold;"><span>Due:</span> <span>${due.toFixed(0)}</span></div>
</div>
  <div style="text-align: center; font-size: 9px; margin-top: 5px;">
    <p style="margin:0;">Thank You!</p>
  </div>
</div>;
}

```

```

function finalizeSale(action) {
  if(cartItems.length === 0) return alert("আপনার কার্ট খালি!");
  let grandTotal =
parseFloat(document.getElementById('bill-grand-total').innerText.replace('Total: ৳ ', '')) || 0;
  let paidAmt = parseFloat(document.getElementById('bill-paid').value) || 0;
  let currentDue = parseFloat(document.getElementById('bill-due-amount').value) || 0;
  let cusName = document.getElementById('cus-search-billing').value.trim() || "Walk-in
Customer";

```

```

let invoiceNo = "INV-" + Math.floor(Math.random() * 90000 + 10000);

```

```

let dateObj = new Date(); let date = dateObj.toLocaleString('en-US');
let subT = cartTotal; let disc = parseFloat(document.getElementById('bill-discount').value)
|| 0;
let pDue = document.getElementById('chk-prev-due').checked ? currentCustomerDue : 0;

let customerObj = customersDB.find(c => c.name.toLowerCase() ===
cusName.toLowerCase());
if(!customerObj && cusName.toLowerCase() !== "walk-in customer" &&
cusName.toLowerCase() !== "walk-in") {
    customersDB.push({ name: cusName, phone: "N/A", address: "", category: "Normal",
due: 0, limit: 5000, img: "https://via.placeholder.com/80?text=User" });
    customerObj = customersDB[customersDB.length - 1];
}

let phone = customerObj ? customerObj.phone : "";

// WhatsApp Message Format
let wpMessage = `*${shopName} Digital Memo*\n-----\n*Inv No:*\n
${invoiceNo}\n*Date:* ${date}\n*Customer:* ${cusName}\n\n*Brought Items:*\n`;
    cartItems.forEach(i => { wpMessage += `▪ ${i.name} - ${i.qty} x ₳${i.price} = ₳${(i.qty *
i.price).toFixed(2)}\n`; });
    wpMessage += `-----\n*Subtotal:* ₳${subT.toFixed(2)}`; if(disc > 0)
wpMessage += ` \n*Discount:* -₳${disc.toFixed(2)}`; if(pDue > 0) wpMessage += ` \n*Prev
Due:* +₳${pDue.toFixed(2)}`;
    wpMessage += ` \n*Total Bill:* ₳${grandTotal.toFixed(2)}\n*Paid Amount:*
₳${paidAmt.toFixed(2)}\n*Due Amount:* ₳${currentDue.toFixed(2)}\n\n_Thank you for
shopping!_`;

// Handle Actions (Print, WhatsApp, Save)
if(action === 'print') {
    const printArea = document.getElementById('print-area');
    const selectedSize = document.getElementById('setting-print-size') ?
document.getElementById('setting-print-size').value : 'thermal';

    if(selectedSize === 'thermal') {
        printArea.className = 'thermal-print';
    } else if(selectedSize === 'a4-landscape') {
        printArea.className = 'a4-print landscape';
    } else {
        printArea.className = 'a4-print';
    }
}

printArea.innerHTML = generateInvoice(invoiceNo, date, cusName, subT, disc, pDue,
grandTotal, paidAmt, currentDue, cartItems);
setTimeout(() => { window.print(); }, 500);
} else if (action === 'whatsapp') {
    if(!phone || phone === "N/A" || cusName.toLowerCase() === "walk-in customer")
alert("সঠিক মোবাইল নম্বর নেই! শুধু সেল সেভ হচ্ছে।");
}

```



```

        else window.open(`https://wa.me/${phone.startsWith('0') ? '88' + phone :
phone}?text=${encodeURIComponent(wpMessage)}`, '_blank');
    } else { alert(` Invoice ${invoiceNo} Saved!`); }

    // Update DB Data
    cartItems.forEach(i => { let p = productsDB.find(prod => prod.name === i.name); if(p)
p.stock -= i.qty; });
    salesHistoryDB.unshift({ invoice: invoiceNo, date: date, rawDate: dateObj, customer:
cusName, total: grandTotal, paid: paidAmt, due: currentDue });

    if(customerObj) {
        customerObj.due = (customerObj.due - pDue) + currentDue;
        if(customerObj.due < 0) customerObj.due = 0;

        if(customerObj.limit && customerObj.due > customerObj.limit) {
            alert('⚠️ সতর্কবার্তা: ${customerObj.name} এর বকেয়া তার লিমিট (৳${customerObj.limit})
ক্রস করেছে!');
        }
    }

    saveData();
    renderAllTables(); populateDropdowns(); renderSalesHistory();
    cartItems = []; renderCart(); document.getElementById('chk-prev-due').checked = false;
    document.getElementById('bill-discount').value = "";
    walkIn();
}

// ===== 10. SMART DUE LIST & SALES =====
function renderDueTable() {
    const tbody = document.getElementById('due-list-body');
    const searchTerm = document.getElementById('due-search') ?
document.getElementById('due-search').value.toLowerCase() : "";
    const sortBy = document.getElementById('due-sort') ?
document.getElementById('due-sort').value : 'high';
    if(!tbody) return;

    let debtors = customersDB.filter(c => c.due > 0 &&
(c.name.toLowerCase().includes(searchTerm) || c.phone.includes(searchTerm)));

    if(sortBy === 'high') debtors.sort((a, b) => b.due - a.due);
    else if(sortBy === 'name') debtors.sort((a, b) => a.name.localeCompare(b.name));
    else if(sortBy === 'old') debtors.sort((a, b) => b.due - a.due);

    tbody.innerHTML = ""; let totalDue = 0;

    debtors.forEach(c => {
        totalDue += c.due;
        let isOld = c.due > (c.limit || 5000);

```

```

let alertTag = isOld ? '<span title="High Alert" style="cursor:help;">⚠️</span>' : '';

tbody.innerHTML += `
  <tr>
    <td onclick="viewCustomerLedger('${c.name}')" style="cursor:pointer;">
      <div style="display:flex; align-items:center; gap:10px;">
        
        <div><b>${c.name} ${alertTag}</b><br><small
style="color:#64748b;">${c.phone}</small></div>
      </div>
    </td>
    <td style="font-size: 12px; color: #64748b;">Active Ledger</td>
    <td style="color:#ef4444; font-weight:bold; font-size: 16px;">৳
${c.due.toFixed(2)}</td>
    <td class="no-print" style="text-align: center;">
      <div style="display:flex; gap:5px; justify-content:center;">
        <button class="btn btn-success btn-small"
onclick="sendDueWhatsApp('${c.phone}', '${c.name}', ${c.due})" title="WhatsApp Msg"><i
class="fa-brands fa-whatsapp"></i></button>
        <a href="tel:${c.phone}" class="btn btn-primary btn-small" title="Call"><i
class="fa-solid fa-phone"></i></a>
        <button class="btn btn-warning btn-small"
onclick="viewCustomerLedger('${c.name}')" title="Collect Money">৳ Collect</button>
      </div>
    </td>
  </tr>`;
});

if(document.getElementById('total-market-due-display'))
document.getElementById('total-market-due-display').innerText = `৳ ${totalDue.toFixed(2)}`;
if(document.getElementById('total-debtors-display'))
document.getElementById('total-debtors-display').innerText = `${debtors.length} জন`;
}

function sendDueWhatsApp(phone, name, amount) {
  if(!phone || phone === "N/A") return alert("মোবাইল নম্বর নেই!");
  let msg = `আসসালামু আলাইকুম ${name}, আপনার কাছে দোকানের ${amount.toFixed(2)} টাকা বাকি
আছে। অনুগ্রহ করে দ্রুত পরিশোধ করুন। ধন্যবাদ।`;
  window.open(`https://wa.me/${phone.startsWith('0')} ? '88'+phone :
phone}?text=${encodeURIComponent(msg)}`, '_blank');
}

function renderSalesHistory(filterCus = "All") {
  const tbody = document.getElementById('sales-history-body'); if(!tbody) return;
  tbody.innerHTML = ""; let totalS = 0, totalP = 0, totalD = 0;

  const filterSelect = document.getElementById('sales-customer-filter');

```

```

    if(filterSelect && filterSelect.options.length === 1) {
        customersDB.forEach(c => { let opt = document.createElement('option'); opt.value =
c.name; opt.text = c.name; filterSelect.add(opt); });
    }

    salesHistoryDB.forEach(sale => {
        if(filterCus === "All" || sale.customer === filterCus) {
            totalS += sale.total; totalP += sale.paid; totalD += sale.due;
            tbody.innerHTML += `<tr><td style="color:#3b82f6;
font-weight:bold;">${sale.invoice}</td><td>${sale.date}</td><td><b>${sale.customer}</b></td><td>
<b>${sale.total.toFixed(2)}</td><td style="color:#10b981;">
<b>${sale.paid.toFixed(2)}</td><td style="color:#ef4444;">
<b>${sale.due.toFixed(2)}</td><td class="no-print"><button class="btn btn-outline btn-small"
onclick="printReport('sales-print-area')"><i class="fa-solid fa-print"></i></button></td></tr>`;
        }
    });
    if(document.getElementById('filter-total-sell'))
document.getElementById('filter-total-sell').innerText = `<b>${totalS.toFixed(2)}</b>`;
    if(document.getElementById('filter-total-paid'))
document.getElementById('filter-total-paid').innerText = `<b>${totalP.toFixed(2)}</b>`;
    if(document.getElementById('filter-total-due'))
document.getElementById('filter-total-due').innerText = `<b>${totalD.toFixed(2)}</b>`;
}
function filterSalesHistory() {
renderSalesHistory(document.getElementById('sales-customer-filter').value); }

// ===== 11. COST, DASHBOARD & ANALYTICS LOGIC
=====
function switchCostTab(type) {
    currentCostTab = type;

    document.getElementById('btn-daily-cost').classList.replace(type==='daily'? 'btn-outline': 'btn-
primary', type==='daily'? 'btn-primary': 'btn-outline');

    document.getElementById('btn-monthly-cost').classList.replace(type==='monthly'? 'btn-outlin
e': 'btn-primary', type==='monthly'? 'btn-primary': 'btn-outline');
    document.getElementById('exp-cat-daily').classList.toggle('hidden', type!=='daily');
    document.getElementById('exp-cat-monthly').classList.toggle('hidden', type!=='monthly');
    document.getElementById('cost-type-label').innerText = type === 'daily' ? "Daily" :
"Monthly";
    updateCostUI();
}

function saveCategory() {
    let catName = document.getElementById('newCatName').value; if(!catName) return
alert("ক্যাটাগরির নাম দিন");
    let targetSelect = currentCostTab === 'daily' ? document.getElementById('exp-cat-daily') :
document.getElementById('exp-cat-monthly');

```

```

    let opt = document.createElement('option'); opt.text = catName; targetSelect.add(opt);
    targetSelect.value = catName;
    document.getElementById('addCatModal').classList.add('hidden');
    document.getElementById('newCatName').value="";
}

```

```

function addExpense() {
    const cat = currentCostTab === 'daily' ? document.getElementById('exp-cat-daily').value :
    document.getElementById('exp-cat-monthly').value;
    const amount = parseFloat(document.getElementById('exp-amount').value) || 0;
    if (amount === 0) return alert("টাকার পরিমাণ দিন!");
    expensesDB.unshift({ id: Date.now(), type: currentCostTab, cat: cat, amount: amount,
    date: new Date().toLocaleString() });
    saveData(); document.getElementById('exp-amount').value = "";
    document.getElementById('exp-note').value = "";
    updateCostUI();
}

```

```

function deleteExpense(id) { expensesDB = expensesDB.filter(e => e.id !== id); saveData();
updateCostUI(); }

```

```

function updateCostUI() {
    const costList = document.getElementById('exp-list-body'); if(!costList) return;
    costList.innerHTML = ""; let total = 0;
    expensesDB.filter(e => e.type === currentCostTab).forEach(e => {
        total += e.amount;
        costList.innerHTML += `<tr><td><b>${e.cat}</b> <span style="font-size:10px;
color:#64748b;">{e.date}</span></td><td style="color: #ef4444; font-weight: bold;">৳
${e.amount.toFixed(2)}</td><td><button class="btn btn-danger btn-small"
onclick="deleteExpense(${e.id})"><i class="fa-solid fa-trash"></i></button></td></tr>`;
    });
    document.getElementById('total-exp-display').innerText = `৳ ${total.toFixed(2)}`;
}

```

```

function switchDashboardView(viewMode) {
    let now = new Date();
    let tSell = 0, tProfit = 0, tDueCol = 0, tCost = 0, countBills = 0;
    let mSell = 0, mProfit = 0;

    salesHistoryDB.forEach(sale => {
        let sd = new Date(sale.rawDate || sale.date);
        let profitAmt = (sale.total - (sale.total * 0.85)); // Demo 15% profit
        if(sd.getMonth() === now.getMonth() && sd.getFullYear() === now.getFullYear()) {
            mSell += sale.total; mProfit += profitAmt; }
        if(sd.toDateString() === now.toDateString()) { tSell += sale.total; tProfit += profitAmt;
            countBills++; }
    });
}

```

```

dueCollectionHistoryDB.forEach(col => {
  let cd = new Date(col.rawDate || col.date);
  if(cd.toString() === now.toString()) tDueCol += col.amount;
});

expensesDB.forEach(exp => {
  let ed = new Date(exp.date);
  if(ed.toString() === now.toString()) tCost += exp.amount;
});

let totalMarketDue = customersDB.reduce((sum, cus) => sum + cus.due, 0);

if(document.getElementById('dash-today-profit'))
document.getElementById('dash-today-profit').innerText = `₹ ${tProfit.toFixed(2)}`;
if(document.getElementById('dash-today-sell'))
document.getElementById('dash-today-sell').innerText = `₹ ${tSell.toFixed(2)}`;
if(document.getElementById('dash-today-cost'))
document.getElementById('dash-today-cost').innerText = `₹ ${tCost.toFixed(2)}`;
if(document.getElementById('dash-today-due-col'))
document.getElementById('dash-today-due-col').innerText = `₹ ${tDueCol.toFixed(2)}`;
if(document.getElementById('dash-today-cash'))
document.getElementById('dash-today-cash').innerText = `₹ ${((tSell + tDueCol) -
tCost).toFixed(2)}`;
if(document.getElementById('dash-today-bills'))
document.getElementById('dash-today-bills').innerText = `${countBills} ₹`;
if(document.getElementById('dash-today-return'))
document.getElementById('dash-today-return').innerText = `₹ 0`;

if(document.getElementById('dash-month-profit'))
document.getElementById('dash-month-profit').innerText = `₹ ${mProfit.toFixed(2)}`;
if(document.getElementById('dash-month-sell'))
document.getElementById('dash-month-sell').innerText = `₹ ${mSell.toFixed(2)}`;
if(document.getElementById('dash-total-due'))
document.getElementById('dash-total-due').innerText = `₹ ${totalMarketDue.toFixed(2)}`;

updateChartData(); updateDashboardAlerts();
}

function updateDashboardAlerts() {
  let transHtml = "";
  salesHistoryDB.slice(0, 5).forEach(s => { transHtml +=
`<li><div><b>${s.customer}</b><br><small
style="font-size:10px;">${s.date}</small></div><b style="color:#10b981;">+
₹${s.total.toFixed(0)}</b></li>`; });
  document.getElementById('dash-recent-trans').innerHTML = transHtml || '<li>No
transactions</li>';

  let dueHtml = "";

```

```

    let topDebtors = [...customersDB].filter(c => c.due > 0).sort((a, b) => b.due -
a.due).slice(0, 5);
    topDebtors.forEach(c => { dueHtml += `<li><span><i class="fa-solid fa-user"></i>
${c.name}</span><b style="color:#ef4444;">₮${c.due.toFixed(0)}</b></li>`; });
    document.getElementById('dash-due-reminder').innerHTML = dueHtml || '<li>All
clear!</li>';

    let stockHtml = "";
    let lowStocks = [...productsDB].filter(p => p.stock <= 5).slice(0, 5);
    lowStocks.forEach(p => { stockHtml += `<li><span><i class="fa-solid fa-box"></i>
${p.name}</span><b style="color:#ef4444;">₮${p.stock} ${p.unit||'Pcs'}</b></li>`; });
    document.getElementById('dash-low-stock').innerHTML = stockHtml || '<li>Stock is
healthy!</li>';

    let topProdHtml = "";
    [...productsDB].filter(p=>p.isTop).slice(0,5).forEach(p => { topProdHtml += `<li><span>🔥
${p.name}</span><b style="color:#3b82f6;">₮${p.sell}</b></li>`; });
    document.getElementById('dash-top-products').innerHTML = topProdHtml || '<li>No top
products</li>';
}

function initChart() {
    const ctx1 = document.getElementById('myChart');
    let config = { type: 'line', data: { labels: ['6 Days Ago','5 Days Ago','4 Days Ago','3 Days
Ago','2 Days Ago','Yesterday','Today'], datasets: [{ label: 'Sales Trend (₮)', data:
[0,0,0,0,0,0,0], borderColor: '#3b82f6', backgroundColor: 'rgba(59, 130, 246, 0.2)',
borderWidth: 2, fill: true, tension: 0.4 } ] }, options: { responsive: true, maintainAspectRatio:
false } };
    if(ctx1 && typeof Chart !== 'undefined') salesChart = new Chart(ctx1, config);
}

function updateChartData() {
    if(!salesChart) return;
    let dayTotals = [0,0,0,0,0,0,0]; let labels = [];
    for(let i=6; i>=0; i--) {
        let d = new Date(); d.setDate(d.getDate() - i);
        labels.push(i===0?'Today':i===1?'Yesterday':d.toLocaleDateString('en-US',
{weekday:'short'}));
        salesHistoryDB.forEach(s => {
            let sd = new Date(s.rawDate || s.date);
            if(sd.toDateString() === d.toDateString()) dayTotals[6-i] += s.total;
        });
    }
    salesChart.data.labels = labels;
    salesChart.data.datasets[0].data = dayTotals;
    salesChart.update();
}

```

```

function renderFullAnalytics() {
  document.getElementById('stat-tot-prod').innerText = productsDB.length + " টি";
  document.getElementById('stat-tot-cus').innerText = customersDB.length + " জন";
  let totDue = customersDB.reduce((sum, cus) => sum + cus.due, 0);
  document.getElementById('stat-tot-due').innerText = `₳ ${totDue.toFixed(2)}`;
  let totSale = salesHistoryDB.reduce((sum, sale) => sum + sale.total, 0);
  document.getElementById('stat-tot-sale').innerText = `₳ ${totSale.toFixed(2)}`;
  let totProf = salesHistoryDB.reduce((sum, sale) => sum + (sale.total - (sale.total * 0.85)), 0);
  document.getElementById('stat-tot-prof').innerText = `₳ ${totProf.toFixed(2)}`;
  let totExp = expensesDB.reduce((sum, exp) => sum + exp.amount, 0);
  document.getElementById('stat-exp-ratio').innerText = `₳ ${totExp.toFixed(2)}`;
  document.getElementById('stat-prof-ratio').innerText = `₳ ${totProf.toFixed(2)}`;
  let totalInvestment = productsDB.reduce((sum, p) => sum + (p.stock * p.buy), 0);
  if(document.getElementById('stat-tot-invest'))
  document.getElementById('stat-tot-invest').innerText = `₳ ${totalInvestment.toFixed(2)}`;

  let cusSales = {}; salesHistoryDB.forEach(s => { if(!cusSales[s.customer])
  cusSales[s.customer] = 0; cusSales[s.customer] += s.total; });
  let sortedCus = Object.keys(cusSales).sort((a, b) => cusSales[b] - cusSales[a]).slice(0, 5);
  let cusHtml = "";
  sortedCus.forEach((c, i) => { cusHtml += `<li><b>${i+1}. ${c}</b> <span style="float:right; color:#3b82f6; font-weight:bold;">₳${cusSales[c].toFixed(0)}</span></li>`; });
  document.getElementById('top-customers-list').innerHTML = cusHtml || `<li>কোনো সেল নেই</li>`;

  let topProds = productsDB.filter(p => p.isTop).slice(0, 5); let prodHtml = "";
  topProds.forEach((p, i) => { prodHtml += `<li><b>${i+1}. ${p.name}</b> <span style="float:right; color:#10b981; font-weight:bold;">₳${p.sell}</span></li>`; });
  document.getElementById('top-products-list').innerHTML = prodHtml || `<li>প্রোডাক্ট সিলেক্ট করা নেই</li>`;
  updateAdvChart();
}

```

```

function updateAdvChart() {
  let ctx = document.getElementById('advancedAnalyticsChart'); if(!ctx || typeof Chart === 'undefined') return;
  let labels = ['Jan', 'Feb', 'Mar', 'Apr', 'May', 'Jun', 'Jul', 'Aug', 'Sep', 'Oct', 'Nov', 'Dec'];
  let salesData = new Array(12).fill(0); let profitData = new Array(12).fill(0); let dueData = new Array(12).fill(0);
  salesHistoryDB.forEach(s => { let d = new Date(s.rawDate || s.date); if(d.getFullYear() === new Date().getFullYear()) { salesData[d.getMonth()] += s.total; profitData[d.getMonth()] += (s.total - (s.total*0.85)); dueData[d.getMonth()] += s.due; } });
  if(advAnalyticsChart) advAnalyticsChart.destroy();
  advAnalyticsChart = new Chart(ctx, { type: 'line', data: { labels: labels, datasets: [ { label: 'Sales', data: salesData, borderColor: '#ec4899', backgroundColor: 'rgba(236, 72, 153, 0.1)', borderWidth: 2, fill: true, tension: 0.4 }, { label: 'Profit', data: profitData, borderColor: '#3b82f6', backgroundColor: 'rgba(59, 130, 246, 0.1)', borderWidth: 2, fill: true, tension: 0.4 } ] }, options: { responsive: true, maintainAspectRatio: false } });
}

```

